

Siddalingayya G



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EXPERIENCE SUMMARY

Accomplished industrial work experience of over 20 years in complete phase of Product Development: Project planning, Architecture, Electronics system design, Programming, Design Reviews, End-to-End Validation. Cloud, Data Analytics and IoT, Digital transformation across different domains. Good exposure to Business Strategy making, Project Planning, Execution & Risk management.

Proven strengths include:

- Product development in Automotive technologies : Telematics, CV2X & Automation.
- Good exposure to Cockpit Electronics:Telematics, Infotainment & HMI
- Worked on defining product specifications in Connected Cars, CV2X & Software Defined Vehicles SDV
- Internet of Things (IoT), Data analytics and Cloud Technologies for end-to-end solutions
- Electronics & Embedded Systems: Concept design, Product Development and field testing.
- Rapid Prototyping for Proof-of-Concepts (PoC) and demonstration to the customers
- Good hands on to Data Science/Analytics : ML based Algorithms development for Prognostics

SKILLS SUMMARY

Domain Expertise: New Product Development (NPD), Electronics Controller Units, Electronics Systems Design, Internet of Things (IoT), Mob Networks, Data Analytics.

Artificial Intelligence: AI & applications across Product Development & Program Management.

Automation: Entire E2E test case scenarios using base station simulator for 4G,5G RF & GNSS

Tools: Vector Automotive, Doors, JIRA, Doors & Program Management

Cloud Service: Amazon Web Services (AWS) and Google Cloud Platform (GCP)

Patent & Technical Publications:

- Granted Patent - No. IN564183 A SYSTEM FOR MONITORING ROAD SURFACE CONDITION AND A METHOD THEREOF (Granted date: 28-03-2025)
- "Smart Urban Rooftop Greenhouse with WSN Based Precision Technology"
Journal of Advanced Agricultural Technologies, Vol. 6, No. 3, pp. 171-174, September 2019
- Non-Invasive Real Time Error State Detection For Tractors Using Smart Phone Sensors & Machine Learning, by SAE International in United States, January, 2019
- Soil & Silicon: Exploitation to Sustainability, Conference at IIT Madras, June 2016

Technical Membership and Academic Merits:

- Secured All India Rank of 271, in Graduate Aptitude Test in Engineering (GATE) 2004, qualified with 94.87 percentile in Electronics & Instrumentation.
- Secretary: IEEE Nano Technology, Bangalore Chapter (2015-2017)
- Associate Member: The Indus Entrepreneurs (TiE), Bangalore (2016-2018)
- Panel Member: Special Interest Group (SIG) for Startups, IoT Forum & TiE Bangalore

CURRENT AND PREVIOUS EMPLOYMENT DETAILS:

Employer	Location	From	To	Designation
Mahindra & Mahindra	Chennai	Feb-2018	Present	Principal Engineer & Dept Head : System Development, Automotive
Precision Farmers-Founder	Bangalore	Jan-2015	Feb-2018	Founder/Entrepreneur
Rossell Techsys	Bangalore	Apr-2011	Dec-2014	Lead Engineer
Wifinity Technologies	Bangalore	Jan-2009	Mar-2011	Sr. Design Engineer
QuEST Global	Bangalore	Apr-2006	Dec-2008	Sr. Embedded Engineer
Indian Institute of Science (IISc)	Bangalore	Jul-2004	Apr-2006	Project Engineer
BPL Medical Technologies	Bangalore	Feb-2002	Jun-2004	Engineer

CONNECTED CARS : TELEMATICS CONTROL UNIT, INFOTAINMENT & CV2X: PROJECTS EXECUTED

Current: Mahindra & Mahindra, Chennai (8 years: Feb 2018 -Present)

Principal Engineer & Dept Head : System Development, Vehicle Electronics & Software
Mahindra Research Valley (MRV), Mahindra & Mahindra Ltd, Chennai

Role: Heading a COE for Connected Technology, Telematics Control Unit & Vehicle to Everything (V2X) towards next generation cars revolutionizing Mobility.

Major Products Delivered:

Born Electric Vehicles (BEV): XEV 9e & BE 6. ICE: XUV700, Scorpio-N, Thar Roxx, XUV3X0, Commercial Vehicles: Bolero Pickup & UPP VEERO.

Key Activities & Responsibilities

Product development using Modem/Network : 2G,3G,5G & CV2X technologies & Cloud Solutions

Embedded SIM enabled solutions for Internet Connectivity

Connected & Smart Features for Vehicle Tracking, Vehicle Health & Status, Safety & Remote Control

Mob App for the Vehicle owners enabling them connected with Vehicle all time.

Integration of Remote Vehicle Diagnostics & Prognostics for preventive maintenance.

Integration of In-car Applications for getting latest News, Food ordering, Weather & Travel planning.

End-to-end Product Functional Verification & Validation, Defects Log & fix by closely interacting with multiple suppliers.

Sr. Lead Engineer, Advanced Technologies-IoT and Data Analytics

Mahindra Research Valley (MRV), Mahindra & Mahindra Ltd, Chennai

Role: New initiative towards Technological intervention for Farm Machinery through Digitization. Be a part of team for developing disruptive solutions and revenue generation, forming new team, strategy and execution. Budget planning, working closely with Finance and Production team for go to market.

IoT for Precision Farming

To develop an integrated (IoT, Weather, Satellite) solution to generate prediction model for selected crops to minimize the input cost and maximize the yield.

Data streams for model development:

- IoT Sensors data collected from the farm
- Data Analytics and Advisory based on
 - Soil nutrients information
 - Satellite Imagery Data (multispectral)
 - Weather data (Rainfall, Temperature, Humidity, Light radiation)

Vehicle Condition Monitoring and Failure Predictions

Develop a methodology for monitoring condition of 'Tractors' and 'Farm Machinery & Implements' in real-time for detecting any deviations from the expected behavior/operation. Develop a Machine Learning Algorithm for predicting the parts failure and Remaining Useful Lifetime (RUL) for critical Tractor components/parts.

Sensors deployed used: Vibration, Load, Pressure, Force and Torque of the Tractors.

Project: IoT based Wireless Sensor Network (WSN) for Smart Farming

Founder/Entrepreneur: Precision Farmers India, Bangalore

Mission: Efficient Fields and Excellent Yields

Role : As an Entrepreneur took complete ownership of Defining problem statement, Requirement specifications, Concept design , Project planning and complete product development. Getting investments for business growth , financial budgeting, firm P/L responsibilities.

Solution Description: Developed an IoT platform for Precision Farming applications. The objective was to monitor environmental variables like Moisture, Humidity, Pressure solar radiation, Temperature, etc... Push the data to cloud server to access through mobile and web app on defined time intervals & event based.

Machine Learning algorithm used for managing Nutrition deficiency and detecting Pest & Diseases at initial stages. Quality is improved as the optimal usage of resources and timely action.

PLM : Product Development SenseSemi Technologies Pvt Ltd, Bangalore

Project: Medical wearable for Personalized Healthcare: ECG, Blood Glucose and Pressure

Duration: (Mar 2017 – Jan 2018)

Description: SenseSmart ® is a personalized health care device for managing chronic health ailments by monitoring vitals: Heart rate, ECG waveform, Blood glucose, Blood pressure and Body temperature.

Physical parameters are stored locally for 7 days and pushed to cloud for historic trend and advice from the Doctors. Product is targeted for FDI qualification for clinical grade and Validation by Practicing doctors

AI and Machine Learning used for predictive Analytics: Heart attack, Diabetes and lifestyle diseases on the data set captured for the individuals.

Major Projects Executed during: 2003-12

- Developing Data Acquisition and Analysis System for GRS Series RADAR
- Development of GSM based Smart Energy meter for 3-Phase AC
- RF communication based Water Flood Detector
- Digital Engine Controller Unit (DECU) Hardware Simulator
- Digital Aero-Engine Controller: Coding & Module Testing as per DO-178B
- PV Solar Energy Meter Firmware Development and Validation.
- Data logger for Quick Lubrication system used in Automobiles
- Tactile sensor for Biomedical application and Electronic Automation

EDUCATION

PGCSMP: Post Graduate Certificate in Senior Management Program
IIM Trichy (2026-27) *Pursuing*

EPGP: Data Analytics and Engineering
Great Lakes Institute of Technologies, Chennai 2018-19

Post Graduation: MTech in Digital Electronics and Communication (*Thesis Pending*)
University: Visvesvaraya Technological University, Karnataka, India 2006-08

Graduation : BE in Electronics and Instrumentation 1997-2001
Institution : Siddaganga Institute of Technology SIT, Bangalore University, INDIA.

Carnatic Vocal & Flute Classical Music : Basic Level 2023-26 Online weekly course.

Areas of Interest:

Smart Mobility, Automobiles, CV2X, 5G Electronics Product Design, Embedded Systems, Wireless Technologies.

Internet of Things (IoT), Data Analytics and Cloud Solutions.

Personal Profile

Date of Birth	: 20th April 1979, 46 years
Passport No.	: P7340943 (Validity 2027)
Father's Name	: Sri Channabasayya G
Marital Status	: Married
Languages Proficient	: English,Hindi, Kannada
Spoken (Basics)	: Sanskrit, Telugu Tamil & Russian
Hobbies	: Carnatic Music, Travel & Books

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